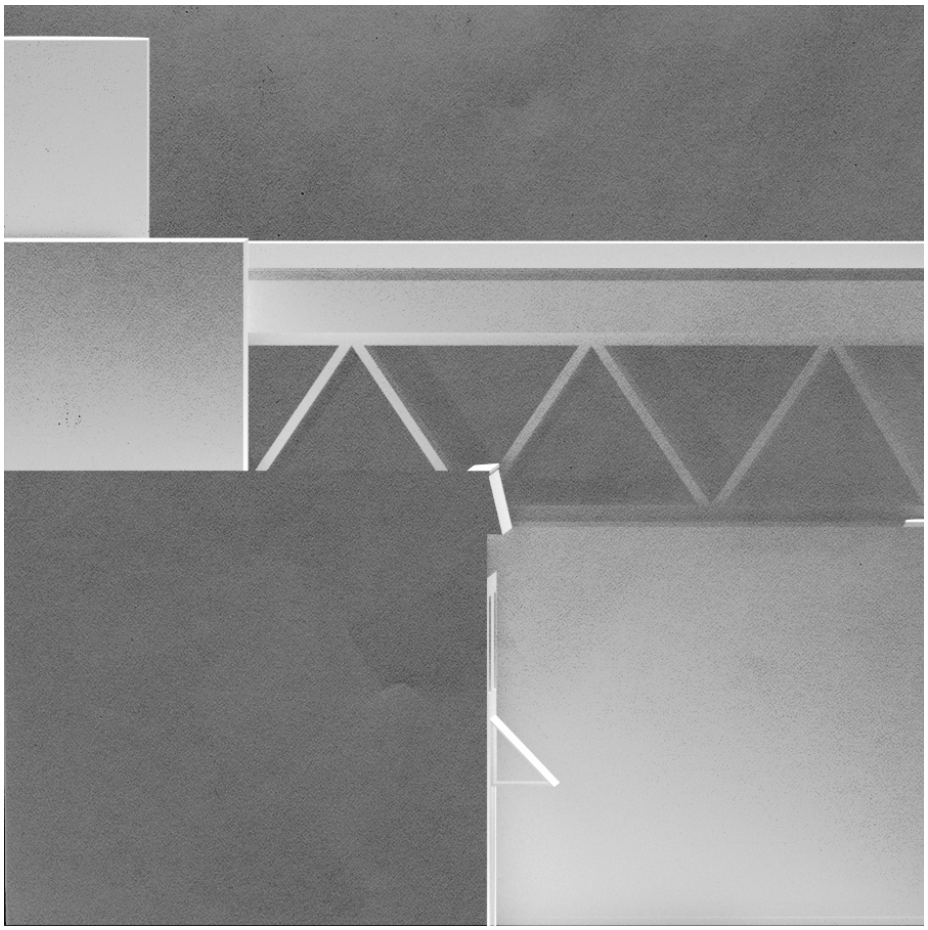


Surplus masses are excavated, unused masses from a construction site. These masses consist of soil, stones, clay and other materials.

In 2021, 12 million tons of earth masses were removed from building sites. Because of local regulations these masses are usually treated as contaminated waste, deposited in landfills.

A recent UN report (Global Resources Outlook 2024) highlights that the growing use of resources like stone and sand is becoming a third major environmental challenge alongside biodiversity loss and climate change.

In times of the ever-growing awareness of our carbon footprint, the linear consumption pattern that we have grown used to, where excavated masses are treated as a waste rather than a resource, is arguably an outdated mindset.

The question can therefore no longer be whether we can build with surplus masses, but rather how we can implement these masses in contemporary construction in a way that is both sustainable, economically viable, and aligned with current building standards and practices.

This has been the focus for this semester, looking into not only how we can build with earth, but also addressing a larger discussion of how earth can be implemented on a large scale in a nordic climate.

In other words; what is the potential in these structures?